

Module 5: Accounting & Financial Management, General Ledger, Multi-Currency Treasury, AP/AR & VAT Compliance

Comprehensive 13-week milestone agreement, engineering effort breakdown, resource cost schedule, and acceptance criteria based on the verified Business Analysis Report **DOC-005 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

DOCUMENT ID	BA REFERENCE DOCUMENT	TOTAL ENGINEERING EFFORT
SL-POP-ERP-MS-005	DOC-005 v1.0 (04-July-2026)	13 Working Weeks (3.25 Months)
FIXED PROJECT COST	TARGET ARCHITECTURE	DOCUMENT STATUS
BD 4,431.818	Multi-Branch Double-Entry Cloud GL	FINAL PROPOSAL

Executive Summary & Master ERP Context

The **Accounting & Financial Management System** (Module 5) serves as the centralized financial intelligence, compliance, and governance backbone of Popular Auto Spare & A/C Parts Co. W.L.L. Unlike standalone accounting software, this system is transaction-driven, inventory-integrated, and profitability-centric. It translates daily operational transactions from procurement, store warehousing, and sales counters into real-time double-entry general ledger entries across all regional branches and international entities.



Core Objective & Business Purpose

To establish a unified, multi-currency double-entry accounting engine with strict two-stage financial posting gates, dynamic landed-cost COGS calculation, 3-way invoice matching for Accounts Payable, risk-controlled Accounts Receivable with automated overdue credit locks, comprehensive Post-Dated Cheque (PDC) treasury management, Bahrain/GCC VAT compliance, and multi-branch consolidation.

Master ERP Transformation Architecture (9 Core Process Modules)

CURRENT SCOPE: MODULE 05
ACTIVE

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting
- 06 Administration & Security
- 07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Mobile Analytics



5-Group Dynamic COA & Double-Entry GL

Hierarchical Assets, Liabilities, Equity, Revenue, and Expense account trees with cost-center mapping and two-stage posting validation.

Double-Entry

Cost Centers

Two-Stage Gate



Treasury, Banking & PDC Management

Multi-bank management, automated bank reconciliation (BRS), Post-Dated Cheque (PDC) clearing lifecycle, and real-time cash flow forecasting.

BRS Recon

PDC Lifecycle

Cash Forecast



Tax, VAT & Branch Setup SOP

GCC/Bahrain VAT return automation, landed-cost COGS tracking, consolidated balance sheet/P&L, and 10-phase new branch setup checklist.

GCC VAT 10%

Landed Cost

Branch SOP

Module 5: Functional Scope & Architecture Breakdown (BA Ref: DOC-005)

Derived directly from the 51 pages of **DOC-005: Accounting & Financial Management Business Analysis Report**, the implementation scope is structured into 10 key architectural domains:

📑 2.1 Dynamic Chart of Accounts & General Ledger (GL)

- **5 Financial Groups:** Assets, Liabilities, Equity, Revenue, and Expenses with multi-level parent-child hierarchy.
- **Multi-Dimensional Coding:** Segmented tagging by Country, Branch, Warehouse, Department, Product Line, and Project.
- **Two-Stage Financial Posting:** Operational drafts (Purchase/Sales/Stock) require finance validation before final GL posting.
- **Journal Management:** Manual Journals, Recurring Entries, Accrual Adjustments, and Reversal / Correction Entries.

💰 2.2 Accounts Payable (AP) & Landed Cost / COGS Engine

- **3-Way Matching:** Automated matching of Purchase Order (PO), Store PVN Goods Receipt, and Supplier Tax Invoice.
- **Landed Cost Apportionment:** Customs duty, marine freight, port clearing, and local transport allocation to unit inventory valuation.
- **AP Sub-Ledger:** Supplier balance tracking, advance payment offsets, payment scheduling, and supplier credit limit exposure.
- **AP Aging & Forecasting:** Automated 30/60/90+ day aging buckets and projected cash outflow schedules.

📄 2.3 Accounts Receivable (AR) & Credit Risk Governance

- **Dual Credit Limits:** Branch-level credit allocation combined with centralized enterprise credit ceiling.

- **Automated Overdue Hard Lock:** Automatic blocking of sales orders and invoices when credit days or limits are breached.
- **Receipt Voucher Management:** Multi-mode collections (Cash, Card, BenefitPay, Cheque, Bank Transfer, Customer Credit Balance).
- **Statement of Accounts (SOA):** Real-time customer ledger generation, aging reports, and automated collection reminders.

2.4 Banking, Treasury & Bank Reconciliation (BRS)

- **Multi-Bank Management:** Dedicated sub-ledgers for operational bank accounts, overdraft facilities, and deposit accounts.
- **Automated Bank Reconciliation:** Electronic statement import, automated transaction matching, and unrepresented cheque tracking.
- **Cash Management:** Branch cash drawer transfers, head office deposits, and petty cash imprest replenishment.
- **Trade Finance & Guarantees:** Tracking of Bank Guarantees (BG), Letters of Credit (LC), and short-term bank financing facilities.

2.5 Post-Dated Cheque (PDC) Management

- **PDC Received (Customer):** Safe custody tracking, maturity alerts, bank deposit slips, clearance, or bounce workflows.
- **PDC Issued (Supplier):** Supplier cheque issuance scheduling, bank balance availability verification, and clearance posting.
- **Bounced Cheque Handling:** Reversal of receipt, customer account re-debiting, bounce penalty logging, and credit lock.
- **PDC Registry & Portfolio:** Centralized dashboard for active, matured, discounted, and cleared cheques.

2.6 Multi-Currency & Foreign Exchange (FX) Engine

- **Multi-Currency Support:** Transactions recorded in original currency (USD, EUR, JPY, AED, SAR, etc.) with functional BHD base.
- **Daily Exchange Rate Matrix:** Automated/manual FX rate feeds with historical rate audit trails.
- **FX Gain/Loss Engine:** Realized gain/loss calculation on settlement and unrealized month-end balance sheet revaluations.

- **Multi-Currency Statements:** Customer and supplier ledger viewing in transaction currency and base BHD.

2.7 Financial Budgeting & Variance Control

- **Multi-Category Budgets:** Revenue, Procurement, Operating Expenses (OPEX), and Capital Expenditure (CAPEX) budgets.
- **Budget Enforcement Rules:** Soft warnings vs Hard blocking when purchase requests exceed allocated departmental budgets.
- **Budget vs Actual Variance:** Real-time variance tracking, utilization percentages, and budget reallocation workflows.
- **Budget Revision Control:** Controlled versioning and approval hierarchy for mid-year budget revisions.

2.8 Tax, VAT & Statutory Compliance Engine

- **GCC / Bahrain VAT Engine:** Automated calculation of standard (10%), zero-rated (0%), and exempt VAT transactions.
- **Input vs Output VAT Reconciliation:** Automated tax ledger reconciliation and periodic VAT return filing sheets.
- **Customs & Import Duties:** Accounting for port customs duties, clearance fees, and reverse-charge mechanism.
- **Tax Audit Trail:** Complete transaction-level tax invoice drill-down for National Bureau for Revenue (NBR) audit readiness.

2.9 Financial Consolidation & Management Reporting

- **Core Statements:** Real-time Balance Sheet, Profit & Loss (Income Statement), Cash Flow Statement, and Trial Balance.
- **Multi-Branch Consolidation:** Consolidated multi-entity financial statements with automated inter-branch eliminations.
- **Trading Profitability Intelligence:** Real-time gross margin and net contribution analysis by SKU, brand, branch, and customer.
- **Executive Financial Dashboard:** Live KPI cockpit (Working Capital, Quick Ratio, DSO, DPO, Cash Runway).

2.10 New Branch Setup SOP & Implementation Guideline

- **10-Phase Standardized SOP:** Business Registration, Org Structure, COA Configuration, Warehouse/Bin Mapping, HR/Payroll, Asset Tagging, Banking, Access Matrix, UAT, and Go-Live.
- **Automated Branch Provisioning:** 1-click generation of branch cost centers, default ledger accounts, and voucher sequences.
- **Inter-Branch Governance:** Standardized SOP for inter-branch transfer pricing, settlement clearing, and head office reporting.

Project Team Structure & Dedicated Resource Pricing

(13 Weeks / 3.25 Months)

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 13-week (3.25 months / 65 working days) delivery of Module 5 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Financial Solutions Architect	BD 545.455	BD 136.364	BD 3.409	3.25 Months (13 Wks)	BD 1,772.727
Back-End Lead Engineer (PHP MVC / Double-Entry APIs)	BD 227.273	BD 56.818	BD 1.420	3.25 Months (13 Wks)	BD 738.636
Front-End Lead Engineer (Accounting UI / React BI)	BD 227.273	BD 56.818	BD 1.420	3.25 Months (13 Wks)	BD 738.636
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.25 Months (13 Wks)	BD 664.773
QA & Financial Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	3.25 Months (13 Wks)	BD 517.045

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Total Dedicated Engineering Team	BD 1,363.636 / mo	BD 340.909 / wk	BD 8.523 / hr	13 Weeks (3.25 Mo)	BD 4,431.818



Fixed Price Guarantee

Total project investment for Module 5 is fixed at **BD 4,431.818**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Detailed Milestone Breakdown & Deliverables (13-Week Roadmap)

The implementation is organized into 5 sequential sprint milestones, ensuring rapid delivery, thorough testing, and seamless integration:

1

Milestone 1: Chart of Accounts (COA) & Double-Entry General Ledger Engine

WEEKS 1 – 3 (3 WKS)

Architecture and implementation of the 5-group Chart of Accounts, multi-branch cost center hierarchy, double-entry GL posting engine, journal management (Manual, Recurring, Adjustment, Reversal), and the two-stage financial posting verification gate.

- ✓ **Hierarchical COA Tree**
5 financial groups, branch/cost-center mapping, and approval change workflow.
- ✓ **Double-Entry GL Posting**
Real-time ledger posting with source module draft validation gates.
- ✓ **Journal Voucher Engine**
JV suite with automated debit/credit balancing and supporting document attachments.

2

Milestone 2: Accounts Payable (AP) & Landed Cost / COGS Valuation Engine

WEEKS 4 – 6 (3 WKS)

Setup of Supplier Master, 3-way invoice matching (PO + Store PVN + Vendor Invoice), freight/customs duty landed cost apportionment, supplier ledger, payment scheduling, advance offsets, and AP aging analysis.

- ✓ **3-Way Invoice Matching**
Automated PO, PVN goods receipt, and vendor invoice reconciliation.

3

✓ **Landed Cost Calculation**

Apportionment of customs duty, freight, and clearing to inventory unit cost.

✓ **AP Sub-Ledger & Aging**

Supplier credit terms, payment forecasting, and 30/60/90+ day aging buckets.

Milestone 3: Accounts Receivable (AR), Credit Limits & Overdue Control

WEEKS 7 – 9 (3 WKS)

Engineering of Customer Credit Profiles, dual credit limits (branch + company), automated overdue blocking rules, receipt vouchers, installment payment plans, and automated Statement of Accounts (SOA).

✓ **Dual Credit Limit Control**

Branch limits combined with centralized corporate credit ceiling enforcement.

✓ **Overdue Customer Lock**

Automatic blocking of sales invoicing upon payment default or credit day breach.

✓ **Receipt Vouchers & SOA**

Multi-mode payment receipts, customer ledger reconciliation, and automated aging reports.

4

Milestone 4: Banking, Treasury, Post-Dated Cheques (PDC) & FX Engine

WEEKS 10 – 11 (2 WKS)

Rollout of multi-bank account management, automated bank reconciliation (BRS), Post-Dated Cheque (PDC) registry (clearing/bounce workflows), multi-currency transaction accounting, and month-end FX revaluations.

✓ **Bank Reconciliation (BRS)**

Statement import, auto-reconciliation, and unpresented cheque tracking.

✓ **PDC Complete Lifecycle**

PDC Received/Issued registers, maturity notifications, and bounced cheque workflows.

✓ **Multi-Currency & FX Engine**

Realized/unrealized FX gain/loss and balance sheet currency revaluations.

Milestone 5: Budgeting, GCC VAT Compliance, Consolidated BI & Branch SOP

WEEKS 12 – 13 (2 WKS)

Implementation of CAPEX/OPEX budget control, Bahrain/GCC VAT return automation, multi-branch consolidated Balance Sheet / P&L, Executive Financial BI dashboard, and the 10-phase new branch setup checklist.

✓ **Budget Variance Control**

OPEX/CAPEX budget limits, soft/hard blocking, and revision workflows.

✓ **GCC VAT Compliance**

Automated 10% VAT return generation and input vs output tax reconciliation.

✓ **Consolidation & Branch SOP**

Multi-entity consolidated financials, elimination entries, and branch setup SOP.

Milestone Payment Structure & Invoicing Schedule (13-Week Model)

Payments are structured in 5 equal milestone disbursements of **20.00% (BD 886.364 each)**, tied strictly to formal milestone deliverables and client sign-off:

Milestone Phase	Deliverable Description	Timeline	Percentage	Amount (BHD)	Payment Trigger & Acceptance Criteria
Milestone 1	Chart of Accounts (COA) & Double-Entry General Ledger Engine	Weeks 1–3 (3 Wks)	20.00%	BD 886.364	Demonstration of 5-group COA to branch cost centers, and double-entry GL posting
Milestone 2	Accounts Payable (AP) & Landed Cost / COGS Engine	Weeks 4–6 (3 Wks)	20.00%	BD 886.364	Sign-off of 3-wk matching (PO+PVN+Invoice) landed cost calculation & AP ledger
Milestone 3	Accounts Receivable (AR), Credit Limits & Overdue Control	Weeks 7–9 (3 Wks)	20.00%	BD 886.364	Verification of credit limit rules automated over customer lock & SOA
Milestone 4	Banking, Treasury, Post-Dated Cheques (PDC) & FX Engine	Weeks 10–11 (2 Wks)	20.00%	BD 886.364	Demonstration of Bank Reconciliation (BRS), PDC clearing/bounce

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
					engine & FX revaluation
Milestone 5	Budgeting, GCC VAT Compliance, Consolidated BI & Branch SOP	Weeks 12–13 (2 Wks)	20.00%	BD 886.364	Delivery of VAT returns, consolidated Balance Sheet/P&L, executive dashboard & final UAT
Total Milestone Project Cost	Complete Module 5 ERP Scope Delivery	13 Weeks	100.00%	BD 4,431.818	Formal Acceptance & Sign-off

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled at the conclusion of each milestone.

15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify deliverables and ledger postings before invoice closure.

Financial SLA & Defect Severity Matrix

Critical GL/Posting Calculation Anomalies (Severity 1) resolved within 4 hours; Major Financial Reporting Defects (Severity 2) within 24 hours; Minor UI adjustments (Severity 3) within 3 business days.

Scope Control & Audit Integrity

Features adhere strictly to the 51-page specifications in DOC-005. All GL journal entries and financial adjustments are permanently logged for statutory external audit compliance.

Stakeholder Authorization & Sign-Off

By digitally endorsing this Milestone & Payment Structure document, all parties confirm adherence to the deliverables, 13-week timeline, and financial schedule defined for **SL-POP-ERP-MS-005**:

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan
Software Architect, SaNDS Lab

✓ PREPARED (15-JUNE-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV
CEO & Managing Director, SaNDS Lab

✓ VERIFIED (04-JULY-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions
Project Management Consultant

🕒 PENDING REVIEW

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts
Authorized Executive Signatory

🕒 PENDING APPROVAL

PENDING SIGNATURE